# Week 13

**Course Level Objectives:**

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

| Module Objectives | Instructional Materials | Learning Activities | Assessments | Tools |
|---|---|---|---|---|
| **1.** Compute the standard error and conservative degrees of freedom for a difference in two independent means. (CLO 5)<br>**2.** Check the conditions for two-sample t procedures. (CLO 5)<br>**3.** Construct and interpret a confidence interval for $\mu_1 - \mu_2$. (CLO 6)<br>**4.** Explain how the order of the groups affects the sign of the difference. (CLO 6)<br>**5.** Perform a two-sample t test, choosing the tail from the research question. (CLO 8) | e-Text: 6.4 Inference for a Difference in Means (6.4-D, 6.4-CI, 6.4-HT)<br>Interactive Lecture: 6.4 Inference for a Difference in Means<br>Slides: 6.4 Inference for a Difference in Means | Participation: Comparing Two Means – video games and GPA CI; Pygmalion-effect test; Your Turn: Exercise and Sleep (two-sample t test and CI) | Online Homework: 6.4 Inference for a Difference in Means (due Sun Nov 22, 11:59 PM)<br><br>Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives)<br><br>Test 3 Study Guide (all objectives)<br><br>Practice Test 3 (optional; due Thu Nov 19) (all objectives) | WileyPLUS<br>StatKey<br>Blackboard<br>Zoom<br>Respondus |
| **6.** Classify a study design as paired or independent samples by checking for a link between observations. (CLO 1, CLO 8)<br>**7.** Define the difference variable d and the order of subtraction. (CLO 8)<br>**8.** Perform a paired t test. (CLO 8)<br>**9.** Construct and interpret a paired t confidence interval. (CLO 6)<br>**10.** Explain why pairing reduces variability. (CLO 8) | e-Text: 6.5 Paired Difference in Means<br>Interactive Lecture: 6.5 Paired Difference in Means<br>Slides: 6.5 Paired Difference in Means | Participation: Paired or Independent? – Your Turn: Paired or Unpaired? (class discussion of study scenarios); define d and the order of subtraction | Online Homework: 6.5 Paired Difference in Means (due Sun Nov 22, 11:59 PM)<br><br>Take-Home Quiz 8 (due Tue Nov 24, 11:59 PM) – covers 6.5 (all objectives)<br><br>Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives)<br><br>Test 3 Study Guide (all objectives)<br><br>Practice Test 3 (optional; due Thu Nov 19) (all objectives) | WileyPLUS<br>StatKey<br>Blackboard<br>Zoom<br>Respondus |

**Also this week:** Last day to withdraw: Thu Nov 19.